

Homework 5

Please read Sections 1.1.2 (through the end of Example 1.4) and 1.3 (through the end of Example 1.10) from Durrett.

I have posted two sets of notes with this assignment to fill in details about the algebra of events and about the rules of probability. Please also read my notes. I do not expect you to memorize what is in those notes, but seeing all these rules of calculating with events and probability at the start will help you.

1. This is a pure counting question—nothing (yet) to do with probability. Suppose we have a team of four people. Call them A, B, C, D. We want to split $\{A, B, C, D\}$ into two groups of two. If the order of the groups matters (maybe group 1 and group 2 have different jobs), then there are $4!/(2! 2!) = 6$ ways to do this. How many ways are there to split the four people into two groups if the order does not matter? (For example, we could split into $\{A, B\}$ and $\{C, D\}$, which is the same as splitting into $\{C, D\}$ and $\{A, B\}$.) List the possibilities.

There are three ways to do this: $\{\{A, B\}, \{C, D\}\}$; $\{\{A, C\}, \{B, D\}\}$; $\{\{A, D\}, \{B, C\}\}$.

A good way to think of “3” is $6 / 2!$. We’ll come back to this idea in a future homework assignment. It will let us count, for example, the number of ways of dividing our class of 15 students into 5 groups of 3, where the order of the groups does not matter.

2. Please work Exercise 17 from Durrett. (Answer from back of book: 0.2; show your method for getting this value using the algebra of events and rules of probability.)

We are given that events A and B are disjoint ($A \cap B = \emptyset$) and that $P(A) = 0.3$ and $P(B) = 0.5$. We are to calculate $P(A^c \cap B^c)$.

We first use the algebra of events (“De Morgan’s laws”) to rewrite $A^c \cap B^c$ as $(A \cup B)^c$. We then have

$$\begin{aligned} P(A^c \cap B^c) &= P((A \cup B)^c) = 1 - P(A \cup B) \text{ by the general rule } P(D^c) = 1 - P(D) \\ &= 1 - (P(A) + P(B)) \text{ since } A \text{ and } B \text{ are disjoint events } = 1 - (0.3 + 0.5) = 0.2. \end{aligned}$$

3. Please work Exercise 21 from Durrett. (Answer from back of book: yes. Convince me by explaining the calculations of $P(A)$, $P(B)$, and $P(A \cap B)$.)

We use the notations from Example 1.2 for outcomes in our sample space Ω . An ordered pair like (3, 2) means that we roll 3 on the red die and 2 on the green die.

$$A = \{(2, 1), (2, 2), (2, 3), (2, 4), (2, 5), (2, 6), (5, 1), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)\}$$

$B = \{(1, 6), (2, 5), (2, 6), (3, 4), (3, 5), (3, 6), (4, 3), (4, 4), (4, 5), (4, 6), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5), (6, 6)\}$

$A \cap B = \{(2, 5), (2, 6), (5, 2), (5, 3), (5, 4), (5, 5), (5, 6)\}$

Since there are 36 outcomes in Ω and all outcomes are equally likely, we have the following probabilities:

$$P(A) = 12 / 36 = 1 / 3$$

$$P(B) = 21 / 36 = 7 / 12$$

$$P(A \cap B) = 7 / 36.$$

Since $P(A) \cdot P(B) = P(A \cap B)$, the events A and B are independent.

4. Please work Exercise 26 from Durrett. Make your reasoning clear so that after reading your solution, I believe your value for $P(B)$ is correct.

We are given that A and B are independent events with $P(A) = 0.4$ and $P(A \cup B) = 0.64$. We are to calculate $P(B)$. Here are two methods to calculate $P(B)$. You just need one!

First method:

We use the formula $P(A \cup B) = P(A) + P(B) - P(A \cap B)$. Since A and B are independent, we have $P(A \cap B) = P(A) \cdot P(B)$, and the formula gives us $0.64 = 0.4 + P(B) - (0.4) P(B)$. Doing a bit of algebra, we find $0.24 = (0.6)P(B)$ and hence $P(B) = 0.4$.

Second method:

From $P(A \cup B) = 0.64$, we conclude $P((A \cup B)^c) = 0.36$. Using the algebra of events (De Morgan's laws), we have $P((A \cup B)^c) = P(A^c \cap B^c)$. Since A and B are independent, so are A^c and B^c ; therefore, $P(A^c \cap B^c) = P(A^c) \cdot P(B^c)$. We have $P(A^c) = 1 - P(A) = 1 - 0.4 = 0.6$. Therefore, solving $0.36 = 0.6 P(B^c)$ for $P(B^c)$, we find $P(B^c) = 0.6$ and hence $P(B) = 0.4$.

5. Please work Exercise 27 from Durrett. (Answer from back of book: 19/27. Make your reasoning clear to convince me that Durrett's probability is correct. Explain, as best you can, using the algebra of events and the rules of probability.)

Let A be the event that the first student solves the problem, B be the event that the second student solves the problem, and C be the event that the third student solves the problem. We want to calculate the probability of the event "at least one student solves the problem." Using the algebra of events, we view this event as $(A^c \cap B^c \cap C^c)^c$. The event inside the parentheses is "no student solves the problem"; the complement of that event is "some student solves the problem." This method is our familiar "at least one" trick.

To finish the problem, we assume that A, B, C are independent events. Durrett often makes the assumptions implicitly. One of our jobs in solving problems is to identify those assumptions and make them when they seem reasonable. In this case, independence makes some sense—you could think of it as telling us that the students are not collaborating on the problem.

Assuming that A, B, C are independent, so are A^c , B^c , and C^c .

We now calculate as follows:

$$P\left(\left(A^c \cap B^c \cap C^c\right)^c\right) = 1 - P\left(A^c \cap B^c \cap C^c\right) \stackrel{!}{=} 1 - P\left(A^c\right)P\left(B^c\right)P\left(C^c\right) \text{ by the independence assumption } \stackrel{!}{=} 1 - \left(\frac{2}{3}\right)\left(\frac{2}{3}\right)\left(\frac{2}{3}\right) \\ \stackrel{!}{=} 1 - \frac{8}{27} \stackrel{!}{=} \frac{19}{27}.$$

6. You may have started to think about the following question in class. It is also Durrett's Exercise 28. Consider three events A, B, C.

- a. Use the algebra of events to write the event "exactly one of A, B, C happens" using $\cup$, $\cap$, and complements. It may help, for example, to think of "only A happens" as "A happens and B does not happen and C does not happen" or, using the algebra of events, as $A \cap B^c \cap C^c$. "Exactly one of A, B, C happens" means "only A happens or only B happens or only C happens."

We can write the event "exactly one of A, B, C happens" as follows:

$$\left(A \cap B^c \cap C^c\right) \cup \left(A^c \cap B \cap C^c\right) \cup \left(A^c \cap B^c \cap C\right).$$

This expression is a disjoint union in the sense that each pair of pieces is disjoint. For example, the first two pieces are disjoint, since any outcome in both pieces is in A and in A^c , and these events are disjoint.

- b. Assume $P(A) = 1/4$, $P(B) = 1/3$, $P(C) = 1/2$ and that the three events are independent. What is the probability that exactly one of the events happens?

Since A, B, C are independent, A , B^c , and C^c are independent; A^c , B, and C^c are independent; and A^c , B^c , and C are independent.

$$P\left(\left(A \cap B^c \cap C^c\right) \cup \left(A^c \cap B \cap C^c\right) \cup \left(A^c \cap B^c \cap C\right)\right) = P\left(A \cap B^c \cap C^c\right) + P\left(A^c \cap B \cap C^c\right) + P\left(A^c \cap B^c \cap C\right) \\ \stackrel{!}{=} \left(\frac{1}{4}\right)\left(\frac{2}{3}\right)\left(\frac{1}{2}\right) + \left(\frac{3}{4}\right)\left(\frac{1}{3}\right)\left(\frac{1}{2}\right) + \left(\frac{3}{4}\right)\left(\frac{2}{3}\right)\left(\frac{1}{2}\right) \text{ by the independence assumption } \stackrel{!}{=} \frac{1}{12} + \frac{1}{8} + \frac{1}{4} \stackrel{!}{=} \frac{11}{24}.$$